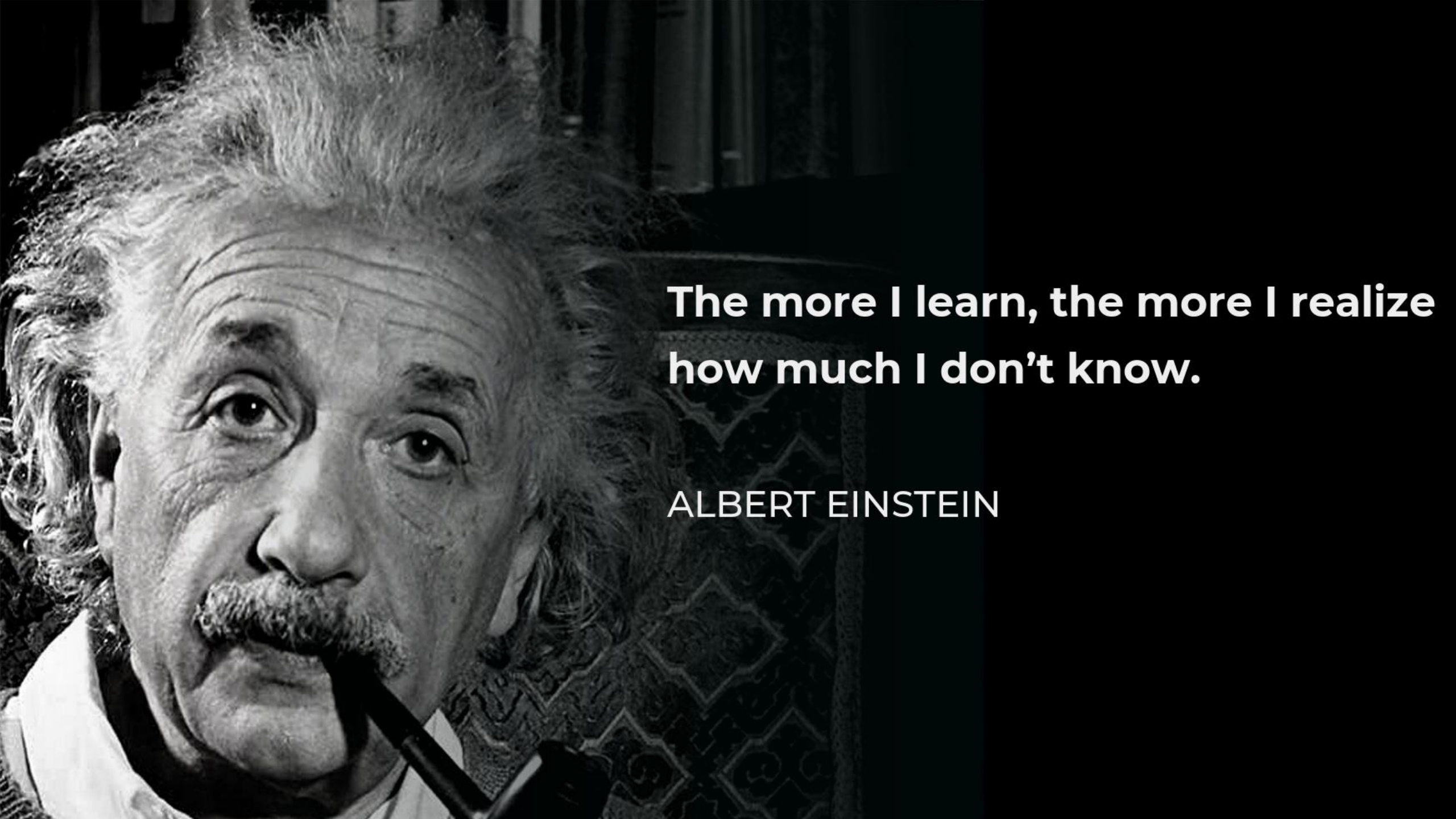

AI for Coders
and Software Engineers
3.5 Days Accelerated Training
Code
Build
Automate
Scale
AI
Mohammad Arshad
Principal Data Scientist | AI Leader
21 Years of Experience
DELL Technologies


The more I learn, the more I realize how much I don’t know.
ALBERT EINSTEIN



Software Engineer
2005 - 2010

absolutdata
-an lirfogain company

Data Scientist
2010 - 2012

DELL Technologies

Data Scientist
2012 - 2014

accenture

Lead Data Scientist
2014 - 2021

MAJID AL FUTTAIM

Data Science Manager
2021 - 2022

DECODING DATA SCIENCE

Founder
2022 - 2025

DELL Technologies

HARVARD BUSINESS REVIEW

Yale

GAP

ICEBREAKER

What do you want to build by Day 4?

What is one AI capability you would love to use confidently, in your own job, by Day 4?

A mixed room

Cybersecurity and OT security, applications, integration, architecture, delivery and plant operations. Different jobs, one shared story.

One journey

Everyone follows the same four days. The scenarios adapt to your role, so the examples always look like your work.

Write your answer down. We come back to it at the end of Day 4.

PREVIOUS BATCHES

People like you have already done this

Photos from earlier cohorts.



IN-APP TRAINING · BATCH 3

Team Use Cases

TEAM 1

Bright Star

Action Tracker Builder Agent

- Builds action trackers from Excel and email inputs
- Captures follow-ups automatically, no manual chasing

TEAM 2

Sarq AI

Control Tower: One Version of Source Truth

- One dashboard, one version of source truth
- Built on synthetic data for a single trusted view

TEAM 3

Sky Net

Predicting Failure Before It Happens

- Predicts failures before they happen
- Holistic dashboard view with MITRE-aligned model security

What Batch 1 built

INSIGHT X

PRISM: Vendor Proposal Review

- Checks vendor proposals against standards
- Review time: 10 days to under 1 day

FORESCASTING 4

Intelligent Workload Manager

- Multi-agent AI on procedures and data
- Faster management decisions

CTRL+ALT+AI

OT Asset Inventory Assistant

- Tracks OT asset vulnerabilities
- Next step: live, real-time data

PGLS

Third-Party Risk & Compliance Agent

- AI risk assessment for suppliers
- Covers both risk and compliance

Every project keeps a person making the final decision. By Day 4, your team adds the next one.



What Batch 2 built

DATA IDEATOR

RFP Evaluation Agent

- Scores proposals against requirements
- Target: 75% less review effort

MANGROVE

Project Pulse: Portfolio Governance

- One view of projects, risks and OKRs
- Across the whole CIO function

TEAM 404

ARC: Asset Risk Intelligence Center

- Links sensors, work orders, inspections
- Surfaces new asset risks, with evidence

FALCONS

Equipment Maintenance AI Assistant

- Faster fault diagnosis for technicians
- Parts, procedures and people lined up

Every project keeps a person making the final decision. By Day 4, your team adds the next one.




AI POWERED
DATA CLASSIFIER
GROUP-3 (AI RUNNERS)
MUHAMMAD BAIG
MADHAVA GANGANA
MOHAMED AL KHATRI
AYESHA ALRAYSSI
DELL Technologies
ENEC AI &
ML Engineering
ENEC
DELL Technologies
ENEC AI &
ML Engineering
ENEC


Brainstorming
4. Idea Grid
Wild Card
Bright Star
Don't Bother
Low Hanging Fruit
AI

THE LESSON

Start with the decision.

Then choose the data, architecture and technique.

NOT THIS "Where can we put an LLM?"

INSTEAD **"Which decision or risk are we improving?"**

Problem → Value → Data → Architecture → Technique → Application → **Human decision**

The system prepares evidence. People remain accountable.

THE JUDGES

A strong idea needs a workable plan.

The last cohort was scored on both.

IDEA / 10

Relevant · useful · inventive

PLAN / 10

Clear · feasible · practical

TOTAL / 20

A good story is not enough. Show how you will test it.

YOUR TURN

By Day 4, build one decision aid.

Start with your job. Design backward from the human decision.

1 What problem?

2 Whose decision?

3 What value or risk?

4 Which data and documents?

5 Which technique?

6 What evidence and test?

7 Where does a human approve?

Build something useful. Keep the person in control.

IN-APP TRAINING

House Rules

01 Ask Questions

Raise your hand and stop me anywhere. Not clear? Ask me to drill down.

02 Experiment

Try it exactly as shown first, then customize it your own way.

03 Think Capstone

Put what you learn into practice on the project you want to build.

04 Focus on Copilot

Your daily tool. We keep coming back to it for best practices.

DAY 1 · SESSION 1 · 08:30 TO 12:15

Generative AI and the Craft of Prompting

AI for Coders & Software Engineers · ENEC 2026



Mohammad Arshad
Principal Data Scientist | AI Leader
21 Years of Experience



https://github.com/Decoding-Data-Science/enecoct2026

THE FOUR-DAY JOURNEY

From using AI, to building it, to trusting it

One pump, A-001, carries the story. Each day adds one layer.

Day 1

Generative AI and data understanding

Microsoft Copilot

→

Day 2

Build a RAG application on trusted documents

Colab · OpenAI · LlamaIndex

→

Day 3

Agents, tools and intelligent decisions

Colab · OpenAI · LlamaIndex

→

Day 4

Capstone, governance and demo

All of it, plus GitHub

The large language model behind Days 2 to 4 is from OpenAI, called from Python in Google Colab.

WHAT HAPPENS EACH DAY

The week in detail

Day	You learn	You build	Tools
1	Generative AI, prompt engineering, data understanding	Structured prompts and a checked analysis of A-001	Copilot in Excel, Word and chat
2	How LLMs answer from documents (RAG), evaluation	A document assistant that cites its sources	Colab, OpenAI, LlamaIndex
3	Agents, tool calling, a decision tree as a ML tool, governance	An agent that queries data, searches documents and asks a human	Colab, OpenAI, LlamaIndex
4	Integration, governance, business value	A team capstone and a final demo	Colab, OpenAI, GitHub

Today has no code. Day 2 starts with Google Colab and your OpenAI key in Colab Secrets.

TODAY

Today in two sessions

Session 1 · morning

08:30 to 12:15

- The AI ecosystem and how an LLM works
- Where prompt engineering fits
- The seven-part structured prompt
- Practice PE1 to PE4
- Copilot setup and the first exercise

Session 2 · afternoon

13:30 to 16:00

- The case of Pump A-001
- Know the data: tables, documents, security layer
- Live investigation with ten prompts
- Role exercises and a team mini project
- From Copilot to OpenAI: the bridge to Day 2

Working Breaks: 10:30 and 15:00. Lunch: 12:30. The rhythm all day: learn the idea, try it in Copilot, check the answer.

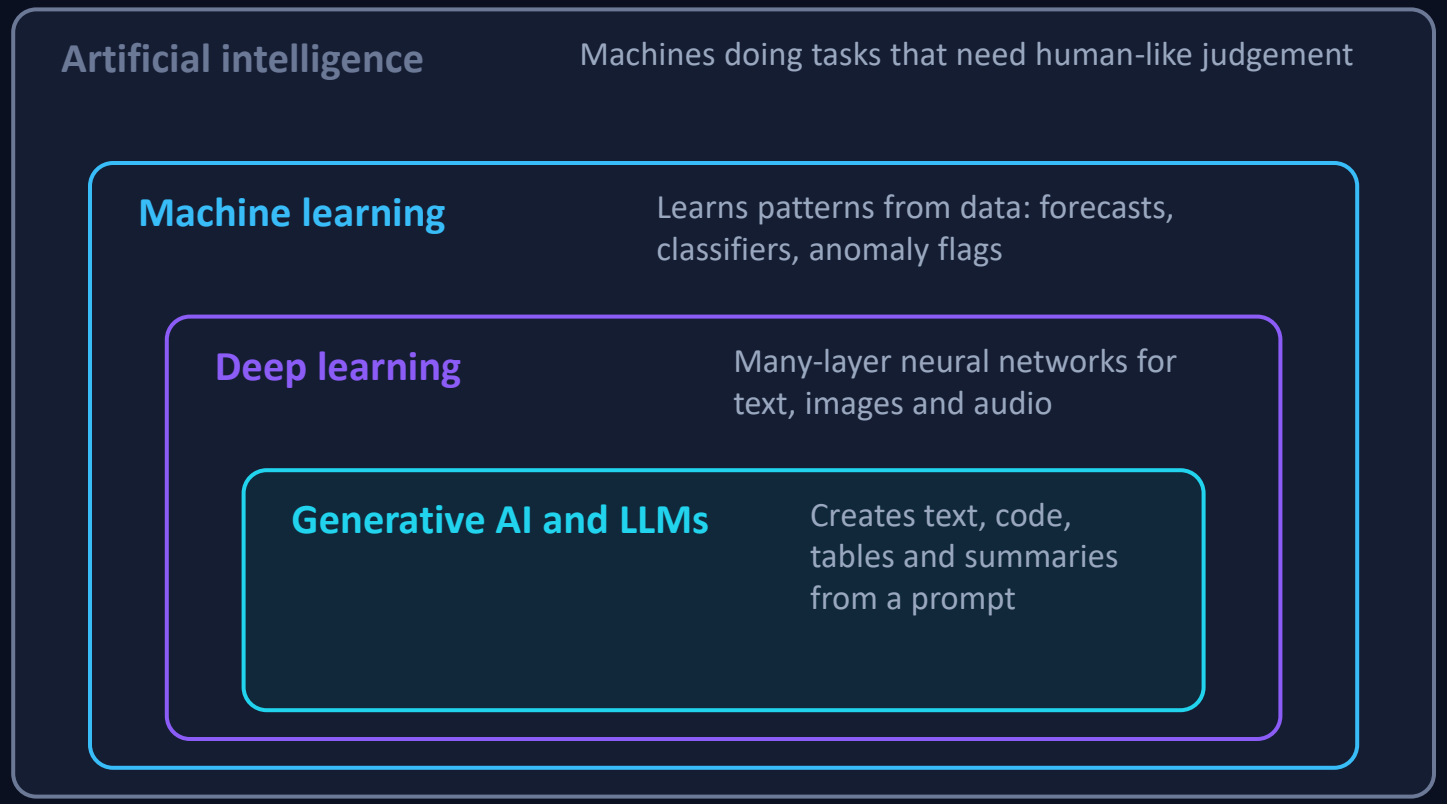
PART 1

The AI ecosystem

THE ECOSYSTEM

AI, machine learning, generative AI: what is what

Nested ideas. Each ring is a smaller, more specific tool.



In this programme

Today and Day 2: generative AI. Day 3: a decision tree, which is classic machine learning, used as a tool by an agent.

Not this week

Training your own large model. We use existing models well, with your data and your rules.

Choose by the question, not by the buzzword.

How an LLM works, in four ideas

Enough to predict when it will help and when it will mislead you.

1 Tokens

Text is cut into small pieces. The model reads and writes tokens, not meaning.

2 Prediction

It predicts the next likely token. Fluent is not the same as true.

3 Context window

It only uses what is in the conversation. If your table is not there, it does not exist for the model.

4 Training cut-off

It knows the public world up to a date. It knows nothing about your plant.

So: give it your evidence, ask for a shape, and check what comes back.

WHERE PROMPT ENGINEERING FITS

The generative AI stack, and the layer you control

Your decision	A qualified person reads the evidence and decides.
Prompt engineering	How you ask: role, task, evidence, constraints, format, check. The layer you control today.
Your data and rules	Tables, documents, approved versions, access rules. This is what makes it about ENEC.
Platforms	Microsoft Copilot (built on OpenAI models) today; the OpenAI API in Colab from Day 2.
Models	Large language models such as OpenAI GPT. Trained by a vendor, used by everyone.

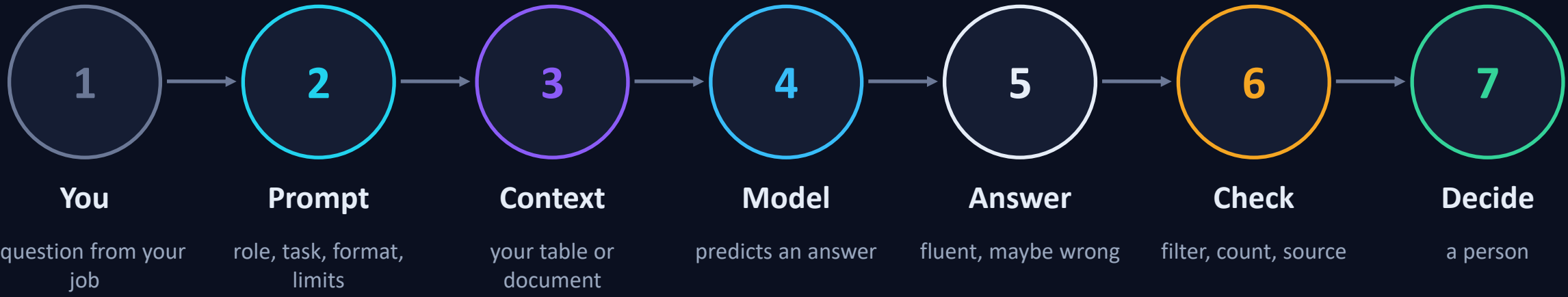
Why it matters

You cannot change the model. You can change what you give it and how you ask. Same model, same data: a better prompt gives a better, checkable answer.

On Day 2 your prompt becomes the system message in code.

ONE REQUEST, END TO END

What happens when you press Enter



You control steps 2, 3, 6 and 7. The model controls only step 4.

Most bad answers come from step 2 or 3: a vague prompt or the wrong evidence.

ENTERPRISE APPLICATIONS

Where LLMs help, and where they need a human

Job to be done	LLM strength	Watch out for
Summarise a long report or incident	Fast and readable	Dropped detail, wrong numbers
Extract fields from documents	Consistent tables	Wrong document version
Explain a table with questions	Quick first look	Silent wrong filter
Draft emails, notes and specifications	Good first draft	Invented facts, wrong tone
Explain code or logs	Plain-language reading	Confident misreading
Decide, approve, authorise	Not its role	A person must own it

Copilot prepares evidence. A qualified person decides.

CHOOSE THE TOOL

Same asset. Different questions. Different tools.

The question	The tool	When
What does this table or document say?	Copilot in Excel, Word and chat	**Day 1**
Answer only from our approved documents	RAG with OpenAI and LlamaIndex	**Day 2**
Gather evidence across tables and documents	An agent with bounded tools	**Day 3**
Is this log window an attack?	A decision tree, used as an agent tool	**Day 3**
Package it as a business solution	Capstone workflow with governance	**Day 4**

Use the simplest tool that answers the question. An agent is the last resort, not the first.

PART 2

Prompt engineering

PROMPT ENGINEERING

A prompt is a small work order

Three things you control decide the answer.

What you ask

Vague question, vague answer. Name the task, the columns and the date.

What it may use

Name the table or the document and its version. Nothing else counts.

How you check

Ask how many rows it used. Compare with a filter you can see.

WEAK

```
"Which assets are risky?"
```

STRUCTURED

```
"Critical assets, patch overdue. A table
with asset_id and exposure. Say how many."
```

A good prompt is one a colleague could act on without asking you a question.

STRUCTURED PROMPTING

The seven-part structured prompt

Short tasks drop parts. Important tasks keep all seven.

Role	Task	Context	Evidence	Constraints	Format	Check
whose view	what to do	why, for whom, as of when	which table or document version	what it must not do	table, bullets, JSON	how many rows, what if missing

A-001 EXAMPLE, COPILOT IN EXCEL

```
Role: You are an OT security analyst.
Task: List Critical assets with Patch_Status Overdue and 2 or more Open_Security_Issues.
Context: Monthly review. Data is as of 2026-08-25.   Evidence: Use only the Asset_360 table.
Constraints: Do not add assets that miss any condition.   Format: Table of asset_id, Network_Exposure,
Open_Security_Issues.
Check: Say how many assets matched. Say "not in the data" if unsure.
```

Expected answer: 4 assets, A-001, A-010, A-065, A-110. Different number? Find out why.

LIVE DEMO · COPILOT IN EXCEL

Watch the same question, asked two ways

ENEC_Asset_360_Security.xlsx · sheet Asset_360

PROMPT A · WEAK

```
Which assets are risky?
```

Prompt A: a vague answer that changes each time you ask.

PROMPT B · STRUCTURED

```
Role: security analyst. Task: list assets that are
Critical AND have an overdue patch. Output: a
table with asset_id, Network_Exposure and
Open_Security_Issues, sorted by
Open_Security_Issues, highest first. Use only this
table. If a column is missing, say so.
```

WHAT SHOULD COME BACK

Prompt B: 7 assets. A-001, A-065, A-010, A-110, A-097, A-098, A-070.

The structure is the skill, not the tool. It works the same in Copilot, ChatGPT or code.

STRUCTURED PROMPTING

Techniques that change the result

Add one at a time. Watch what changes.

1 Be specific

Table, columns, filter, date.

2 Give the format

Table, bullets, JSON keys.

3 Show an example

One or two input and output pairs.

4 Break into steps

Filter, then count, then explain.

5 Ask for evidence

Quote the row or sentence used.

6 Ask for gaps

What can you not conclude?

Then iterate: change one thing, run again, compare. Prompting is a loop, not a spell.

Eight reusable patterns for your job

Pick the pattern for the job, then fill the brackets.

Summarise

For an audience, n bullets, exact numbers.

Extract

Fields into a table, "not stated" if missing.

Compare

Two items, criteria, biggest difference.

Classify

Labels with rules and a reason column.

Explain

No jargon, and what it does not show.

Critique

Against a standard. Gaps, not a rewrite.

Draft

Tone, length, mark the assumptions.

Plan

Steps, owner role, decisions for a person.

Templates for each pattern are in section 5 of the Prompt Engineering Guide.

PRACTICE · 20 MINUTES

Your turn: PE1 to PE4

1. **Open the guide:** Prompt Engineering Guide, section 8: practice prompts PE1 to PE8.
2. **Open the workbook:** ENEC_Asset_360_Security.xlsx from OneDrive, AutoSave on.
3. **Weak first:** Type the weak version of each prompt. Note the answer.
4. **Then structured:** Rewrite it with the seven parts. Run it again.
5. **Check:** Compare both with the check answer in the guide.

Write three lines

1. What Copilot got right.
2. What you had to fix.
3. What it could not know.

We share a few before the break.

No Copilot yet? Pair with a neighbour. Setup comes next if anything fails.

PART 3

Copilot at work

Four jobs we practise today

1 Analysis

Filter, sort and pivot a table. Add columns, charts and formulas. Explain a pattern.

2 Summarisation

Reports, incidents and timelines. Choose the audience and length. List what was left out.

3 Extraction

Fields from a procedure. Tables from long text. Cite the document and version.

4 Productivity

Draft emails and notes. Meeting and review packs. Plans with owner roles.

In every case: name the evidence, ask for a shape, check the result.

MICROSOFT COPILOT

Set up Copilot in Excel so it works first time

Most failures on the day are setup, not prompts.

1

Official account

Sign in with your Microsoft 365 account

2

OneDrive or SharePoint

Save the workbook there, not on your computer

3

AutoSave on

The toggle at the top left

4

.xlsx and a table

Every sheet in the pack is already a table

5

Click inside

Then open Copilot

Know the limits

Large log tables can take 30 seconds or more. Ask narrow questions.

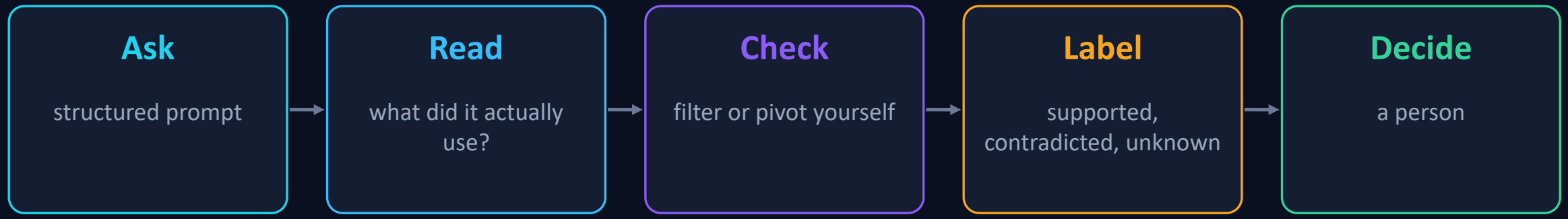
If Copilot is unavailable

Work on a neighbour's screen, or use filters and pivots with the check answers.

VERIFY

Never trust a fluent answer

Every exercise has a check answer. Use it.



Watch for traps

Superseded and draft documents look approved.
Correlation is not cause.
A neat table can hide a wrong filter.

Habits

Ask for the row count and the source.
"Unknown" is a valid answer.
Write down what you corrected.

Copilot prepares evidence. It does not approve a patch or close an incident.

EVERYONE · 10 MINUTES

First exercise: E01, together

ENEC_Asset_360_Security.xlsx · sheet Asset_360

PROMPT

```
Which assets are Critical and have an overdue patch? Show asset_id, Network_Exposure and
Open_Security_Issues, sorted by Open_Security_Issues.
```

Then check it

Filter Asset_Criticality = Critical and Patch_Status = Overdue. Count the rows.

Hands up

Who got 7? Who got a different number? Why?

WHAT SHOULD COME BACK

7 assets: A-001, A-065, A-010, A-110, A-097, A-098, A-070.

LUNCH · BACK AT 13:30

After lunch: the case of Pump A-001

This morning you learned how to ask. This afternoon you investigate a real (synthetic) incident with it.